

Date: 6/8/2023
Task 3: — Importing python modules and packages
in python programming
(a) Weather Report Using datetime.
AIM:

To display the current data and time using python's built-in datetime module in the format:

Algorithm:

1. Import the datetime module.
2. Get the current data and time using datetime.now().
3. Format the data and time using strftime() method with the required format.
4. Display the formatted data and time.

Program:

```
# Weather Report Using datetime  
import datetime
```

```
# Step 1: Get current data and time
```

```
now = datetime.datetime.now()
```

```
# Step 2: Format the output
```

```
formatted = now.strftime('%A, %d %B %Y, %I:%M:%P')
```

```
# Step 3: Display result.
```

```
print("Current data & time: ", formatted)
```

Sample output

Current Date & Time: Tuesday, 12 August 2025, 03:49



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EX NO.	
PERFORMANCE (2)	
NAME (2)	

VEL TECH - CSE	
EX NO.	
PERFORMANCE (5)	
RESULT AND ANALYSIS (5)	
VIVA VOCE (5)	
RECORD (5)	
TOTAL (20)	
SIGN WITH DATE	

RESULT:-

python code for printing the current date and time is completed & executed successfully

Task 3.6 - Create and Use your own module

AIM:

To create a custom python module `myMath.py` containing function `factorial(n)` and `is_prime(n)`, and use them in a main program.

Algorithm:

1. Create `myMath.py`.
 - define `factorial(n)` to return the factorial of n .
 - define `is_prime(n)` to check if n is a prime number.
2. In the main program:
 - Import the `myMath` module.
 - Call `factorial()` and `is_prime()` with sample inputs.
 - Display the results.

Module - `myMath.py`

Python.

`# myMath.py`

```
def factorial(n):
```

```
    fact = 1
```

```
    for i in range(1, n+1):
```

```
        fact *= i
```

```
    return fact
```

```
def is_prime(n):
```

```
    if n <= 1:
```

```
        return False
```

```
    for i in range(2, int(n**0.5)+1):
```

```
        if n % i == 0:
```

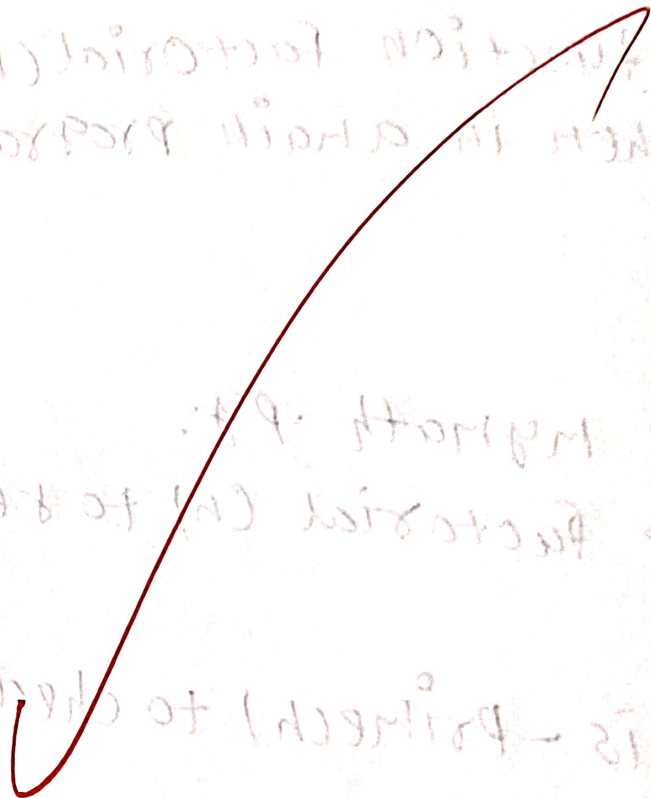
```
            return False
```

```
    return True
```

SampleOut:-

Factorial of 5 = 120

Is 29 Prime? : True.



main program -

```
# main.py
```

```
import mymath
```

```
num = 5
```

```
print(f"Factorial of {num}:", mymath.factorial(num))
```

```
num 2 = 29
```

```
print(f"Is {num} prime?", mymath.is_prime(num))
```

VELTECH - CSE	
EX NO.	
PERFORMANCE (5)	
RESULT AND ANALYSIS (5)	
VIVA VOCE (5)	
RECORD (5)	
TOTAL (20)	
SIGN WITH DATE	

RESULT: Created a custom module's mymath.py and successfully used its function in main program.

TASK 3.(C) Currency converter Using a custom package

AIM:

To create a package currency containing converter.py with a function.
convert (amount, rate) to convert INR to USD.

Algorithm:

1. Create a folder named currency.
2. Inside it, create `__init__.py` (empty) and `converter.py`
3. In `converter.py`, define `convert (amount, rate)`:
 - multiply amount by rate.
 - Return converted value.
4. Import the function in the main program.
5. Convert INR amount to USD

Currency / converter.py

python

```
#currency / converter.py  
def convert (amount, rate):  
    return amount * rate.
```

main.py.

#main.py

```
from currency import converter
```

inr_amount = 1000

usd_rate = 0.012 # Example rate

usd_amount =

```
converter.convert (inr_amount, usd_rate)  
Print (f"int - amount {inr_amount} INR =  
      {usd_amount} USD")
```


Sample Out Pot

$$1000 \text{ INR} = 12.00 \text{ USD}$$



VEL TECH - CSE	
EX NO.	
PERFORMANCE (5)	3
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	18

RESULT: currency converter using a custom package is created & executed successfully

13/8/25

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